For Regular Graphs, Galvin's Kernel Method Works Only on Bipartite Graphs

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Theorem. *Let $k \geq 1$ and let $G = (V, E)$ be a simple k-regular graph. The line graph $L(G)$ has a kernel-perfect orientation with every out-degree at most $k-1$ if and only if G is bipartite.*

"If" is Galvin's theorem [1]. We prove "only if".

Conventions. A *kernel* of a digraph is an independent set K such that every vertex outside K has an out-neighbour in K. $D[S]$ is the induced subdigraph on S. $E(u)$ is the set of edges at u and $N(u)$ the set of neighbours of u. A directed cycle $(x_1, \ldots, x_\ell)$ has $\ell \geq 2$ distinct vertices and arcs $x_i \to x_{i+1}$, indices modulo ℓ; directed walks may repeat vertices. Throughout, $h = \frac{k-1}{2}$.

Four facts about digraphs

Fact 1. *Let T be a tournament on $t \geq 1$ vertices in which every induced subdigraph on a nonempty set has a kernel. Then its vertices can be listed $x_0, \ldots, x_{t-1}$ so that, for $i \neq j$, $x_i \to x_j$ iff $i > j$.*

Proof. Induction on t; $t = 1$ is trivial. Let $t \geq 2$.

(1) T has a kernel K. Any two vertices of T are joined by an arc, so independence gives $|K| \leq 1$. If $K = \varnothing$, a vertex has no out-neighbour in K; so $K = \{s\}$.

(2) By absorption every $x \neq s$ has $x \to s$. Put $x_0 = s$.

(3) Every induced subdigraph of $T - s$ is an induced subdigraph of T, so the induction hypothesis lists $T - s$ as $x_1, \ldots, x_{t-1}$ with $x_i \to x_j$ iff $i > j$.

(4) For $i \geq 1$, $x_i \to x_0$ by (2), and $x_0 \to x_i$ is not an arc since T is a tournament. This agrees with the rule. □

Fact 2. *Let $\ell \geq 3$ be odd and let F be the digraph on distinct $s_1, \ldots, s_\ell$ whose arcs are exactly $s_i \to s_{i+1}$ (indices modulo ℓ). Then F has no kernel.*

Proof. Suppose K is a kernel; let $\kappa_i = 1$ if $s_i \in K$ and $\kappa_i = 0$ otherwise.

(1) The only out-neighbour of s_i is s_{i+1}. So $s_i \notin K$ implies $s_{i+1} \in K$ *[absorption]*, and $s_i \in K$ implies $s_{i+1} \notin K$ *[independence]*.

(2) Hence $\kappa_{i+1} = 1 - \kappa_i \equiv \kappa_i + 1 \pmod 2$ for every i.

(3) Applying (2) ℓ times, $\kappa_1 = \kappa_{1+\ell} \equiv \kappa_1 + \ell$, so ℓ is even. Contradiction. □

Fact 3. *If a digraph has a closed directed walk of odd length, it has a directed cycle of odd length, and that length is at least 3.*

Proof. An odd $\ell \geq 2$ is at least 3. For the first part, strong induction on the odd length L of a closed directed walk $(w_0, \ldots, w_L)$.

(1) $L \neq 1$, since there are no loops. So $L \geq 3$.

(2) If $w_0, \ldots, w_{L-1}$ are distinct, they form a directed cycle of length L.

(3) Otherwise $w_i = w_j$ for some $0 \leq i < j \leq L-1$. Then $(w_i, \ldots, w_j)$ and $(w_0, \ldots, w_i, w_{j+1}, \ldots, w_L)$ are closed directed walks (the second uses the arc $w_i = w_j \to w_{j+1}$), of lengths $j - i$ and $L - (j - i)$, both in $[1, L-1]$.

(4) Their lengths add up to the odd number L, so one is odd and shorter than L; apply the induction hypothesis to it. □

Fact 4. *Let D be a digraph with $d^+(x) = d^-(x)$ for every vertex x, and let $x \to y$ be an arc. Then D has a directed walk from y to x.*

Proof. (1) Let R be the set of vertices reachable from y by a directed walk. Then $y \in R$, and no arc goes from R to outside R.

(2) Let a be the number of arcs inside R and b the number of arcs from outside R into R.

(3) $\sum_{z \in R} d^+(z) = a$ by (1), and $\sum_{z \in R} d^-(z) = a + b$. These sums are equal, so $b = 0$.

(4) The arc $x \to y$ enters R, so it is not counted in $b = 0$. Hence $x \in R$. □

We also use the standard fact that a graph with no odd cycle is bipartite.

Proof of "only if"

Fix a simple k-regular graph $G = (V, E)$ with n vertices, and an orientation D of $L(G)$ that is kernel-perfect and has $d_D^+(e) \leq k - 1$ for every $e \in E$. We show that G is bipartite. Letters e, f denote edges of G (vertices of D); letters u, v, w, a, x, y, z denote vertices of G.

Since G is simple, two distinct edges share at most one end: an edge sharing both ends u, v with e equals $\{u, v\} = e$. This is the only use of simplicity, and it is used in Steps 1 and 2.

Step 1: every vertex ranks its edges

Step 1. *For $u \in V$ and $e \in E(u)$ put $r_u(e) = |\{f \in E(u) : e \to f\}|$, and for $v \in N(u)$ write $r_u(v) = r_u(uv)$. Then:*

(a) $e \mapsto r_u(e)$ is a bijection $E(u) \to \{0, \ldots, k-1\}$, equivalently $v \mapsto r_u(v)$ is a bijection $N(u) \to \{0, \ldots, k-1\}$;

(b) for distinct $e, f \in E(u)$: $e \to f$ iff $r_u(e) > r_u(f)$;

(c) for every edge uv: $d_D^+(uv) = r_u(v) + r_v(u)$.

Proof. (1) $|E(u)| = k \geq 1$ *[k-regular].* Any two edges of $E(u)$ share the end u, so they are adjacent in $L(G)$, and D has exactly one arc between them. So $D[E(u)]$ is a tournament.

(2) Every induced subdigraph of $D[E(u)]$ is an induced subdigraph of D, so it has a kernel *[D kernel-perfect].*

(3) By Fact 1, list $E(u)$ as $f_0, \dots, f_{k-1}$ with $f_i \to f_j$ iff $i > j$. Then f_i points to exactly $f_0, \dots, f_{i-1}$ inside $E(u)$, so $r_u(f_i) = i$. This gives (a), and (b) follows: $f_i \to f_j$ iff $i > j$ iff $r_u(f_i) > r_u(f_j)$.

(4) For (c), let $e = uv$. The neighbours of e in $L(G)$ are the edges of $(E(u) \setminus \{e\}) \cup (E(v) \setminus \{e\})$. These two sets are disjoint: an edge in both would share the two ends u, v with e *[simplicity]*.

(5) The out-neighbours of e in $E(u) \setminus \{e\}$ number $r_u(e) = r_u(v)$, and those in $E(v) \setminus \{e\}$ number $r_v(e) = r_v(u)$ *[definition of r]*. By (4), $d_D^+(e) = r_u(v) + r_v(u)$. □

Step 2: the out-degree budget is used exactly

Step 2. *For every edge uv, $r_u(v) + r_v(u) = k - 1$.*

Proof. (1) Every edge $\{e, f\}$ of $L(G)$ has exactly one common end w *[simplicity]*, and $\{e, f\}$ is then a two-element subset of $E(w)$. Conversely every two-element subset of $E(w)$ is an edge of $L(G)$. So the edges of $L(G)$ correspond bijectively to pairs (w, two-element subset of $E(w)$).

(2) Hence $L(G)$ has $n\binom{k}{2} = \frac{nk}{2}(k-1) = |E|(k-1)$ edges, using $nk = 2|E|$ *[handshake, k-regular]*.

(3) D is an orientation of $L(G)$, so $\sum_{e \in E} d_D^+(e) = |E|(k-1)$.

(4) This is a sum of $|E|$ terms, each at most $k - 1$ *[hypothesis]*. So every term equals $k - 1$.

(5) By Step 1(c), $r_u(v) + r_v(u) = d_D^+(uv) = k - 1$. □

This is the only use of regularity.

Step 3: there is no odd ring

Definition. A *ring* is a cycle $(a_1, \dots, a_\ell)$ of G ($\ell \geq 3$, distinct vertices, indices modulo ℓ) such that $r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1})$ for every i.

Step 3. *G has no ring of odd length.*

Proof. Let $(a_1, \dots, a_\ell)$ be a ring with ℓ odd. Put $e_i = a_i a_{i+1}$ and $S = \{e_1, \dots, e_\ell\}$.

(1) $a_p = a_q$ iff $p \equiv q \pmod{\ell}$. Since $e_i = \{a_i, a_{i+1}\}$ and $e_j = \{a_j, a_{j+1}\}$, a common end of e_i and e_j ($i \not\equiv j$) means $j \equiv i + 1$ or $j \equiv i - 1$, and conversely those pairs share a_{i+1}, respectively a_i.

(2) The e_i are distinct: $e_i = e_j$ with $i \not\equiv j$ would, by the same comparison, force both $j \equiv i - 1$ and $j \equiv i + 1$, so $\ell \mid 2$, contradicting $\ell \geq 3$. So $|S| = \ell$.

(3) By (1), in $L(G)[S]$ the only adjacent pairs are $\{e_{i-1}, e_i\}$, $i = 1, \dots, \ell$, and each such pair occurs for exactly one i (as $\ell \geq 3$). The common end of e_{i-1} and e_i is a_i.

(4) At a_i: $r_{a_i}(e_i) = r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1}) = r_{a_i}(e_{i-1})$ *[ring]*. So the arc of D between them is $e_{i-1} \to e_i$ *[Step 1(b)]*.

(5) By (3) and (4), the arcs of $D[S]$ are exactly $e_1 \to e_2 \to \cdots \to e_\ell \to e_1$.

(6) By Fact 2, $D[S]$ has no kernel. This contradicts kernel-perfectness of D. □

Step 4: a balanced digraph on V

Let O be the digraph on V with an arc $u \to v$ whenever $uv \in E$ and $r_u(v) \le h$. (It has no loops; an edge may receive arcs in both directions.)

Step 4. *(a) Every arc $u \to v$ of O has $uv \in E$.*

(b) For every edge uv, at least one of $u \to v$, $v \to u$ is an arc of O.

(c) If $u \to v$ is an arc of O, then $r_v(u) \ge h$.

(d) $d_O^+(u) = d_O^-(u)$ for every $u \in V$.

Proof. (a) By definition.

(b) If neither, then $r_u(v) > h$ and $r_v(u) > h$, so $r_u(v) + r_v(u) > k-1$, contradicting Step 2.

(c) $r_v(u) = k - 1 - r_u(v) \ge k - 1 - h = h$ *[Step 2, and $r_u(v) \le h$]*.

(d) Let $R = \{0, \dots, k-1\}$, $R_\le = \{r \in R : r \le h\}$, $R_\ge = \{r \in R : r \ge h\}$.

(1) $u \to v$ iff $v \in N(u)$ and $r_u(v) \le h$. Since $v \mapsto r_u(v)$ is a bijection $N(u) \to R$ *[Step 1(a)]*, $d_O^+(u) = |R_\le|$.

(2) $v \to u$ iff $v \in N(u)$ and $r_v(u) \le h$, iff $v \in N(u)$ and $k - 1 - r_u(v) \le h$ *[Step 2]*, iff $v \in N(u)$ and $r_u(v) \ge h$. By the same bijection, $d_O^-(u) = |R_\ge|$.

(3) $\phi(r) = k - 1 - r$ is an involution of R, and $r \le h$ iff $\phi(r) \ge h$. So ϕ maps $R_\le$ bijectively onto $R_\ge$, and $|R_\le| = |R_\ge|$. □

Step 5: directed cycles of O are rings

Step 5. *Every directed cycle $(a_1, \dots, a_\ell)$ of O with $\ell \ge 3$ is a cycle of G and a ring.*

Proof. (1) The a_i are distinct, $\ell \ge 3$, and $a_i a_{i+1} \in E$ *[Step 4(a)]*. So $(a_1, \dots, a_\ell)$ is a cycle of G.

(2) Fix i. The arc $a_i \to a_{i+1}$ gives $r_{a_i}(a_{i+1}) \le h$ *[definition of O]*. The arc $a_{i-1} \to a_i$ gives $r_{a_i}(a_{i-1}) \ge h$ *[Step 4(c)]*.

(3) $a_{i+1} \ne a_{i-1}$ (distinct vertices, $\ell \ge 3$), so $r_{a_i}(a_{i+1}) \ne r_{a_i}(a_{i-1})$ *[Step 1(a)]*.

(4) By (2) and (3), $r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1})$. This holds for every i, so the cycle is a ring. □

Step 6: O has no odd closed directed walk

Step 6. *Every closed directed walk of O has even length.*

Proof. If one had odd length, O would have a directed cycle of odd length $\ell \ge 3$ *[Fact 3]*, which is a ring of G of odd length *[Step 5]*, contradicting Step 3. □

Step 7: every arc of O can be walked back, in an odd number of steps

Step 7. *If $x \to y$ is an arc of O, then O has a directed walk from y to x, and every such walk has odd length.*

Proof. (1) A walk P from y to x exists *[Step 4(d) and Fact 4]*.

(2) For any such P, the arc $x \to y$ followed by P is a closed directed walk of length $1 + |P|$. It is even *[Step 6]*, so $|P|$ is odd. □

Step 8: every cycle of G is even

Step 8. *Every cycle of G has even length.*

Proof. Let $(z_1, \ldots, z_\ell)$ be a cycle of G, indices modulo ℓ.

(1) For each i, $z_i \to z_{i+1}$ or $z_{i+1} \to z_i$ is an arc of O *[Step 4(b)]*.

(2) If $z_i \to z_{i+1}$ is an arc, let $W_i = (z_i, z_{i+1})$, of length 1.

(3) Otherwise $z_{i+1} \to z_i$ is an arc, and Step 7 (with $x = z_{i+1}$, $y = z_i$) gives a directed walk W_i from z_i to z_{i+1} of odd length.

(4) The concatenation $W = W_1 W_2 \cdots W_\ell$ is a closed directed walk of O (from z_1 back to $z_{\ell+1} = z_1$) of length $|W_1| + \cdots + |W_\ell|$.

(5) A sum of ℓ odd integers is congruent to ℓ modulo 2, so $|W| \equiv \ell$.

(6) $|W|$ is even *[Step 6]*. So ℓ is even. □

Conclusion

By Step 8, G has no odd cycle, so G is bipartite. □

References

[1] F. Galvin, The list chromatic index of a bipartite multigraph, *Journal of Combinatorial Theory, Series B* 63 (1995), pages 153 to 158.